HLT Diphoton Trigger Efficiency — 2024 Low-Mass

First results: tag-and-probe, η–R_9 reweighted standard (SS) and fullPath selections

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CMS low-mass $H \to \gamma\gamma$

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Results repo: HLT-diphoton-efficiency-2024-results · Interactive explorer: GitHub Pages

Overview

- **Measurement:** tag-and-probe (TnP) efficiency of the *two legs* of the CMS diphoton high-level-trigger path, on **2024 data**, using $Z \to ee$ electrons as photon probes.
- **Why reweight:** $Z \to ee$ probe photons have different $(R_9, |\eta|)$ distributions than $H \to \gamma\gamma$ signal photons, so each probe is reweighted in $(R_9, |\eta|)$ to resemble the $H \to \gamma\gamma$ population *before* the efficiency fit.
- **2024 scope** — one data-taking era, crossed along two axes:
  - η–R_9 reweighting derived `noPresel` / `withPresel` (without / with the photon preselection);
  - trigger selection: standard track-isolation (`SS`) / `fullPath` (`probe_passFullPath`$= 1$ on top).
- Two legs (*seeded* / *unseeded*), five η/R_9 categories, probe-E_T bins.
- **Deliverables:** public results repo, a production/verification PDF, and an interactive cross-year explorer (2022/2023/2024).

Shown here: the η–R_9-reweighted result (reweighting derived without preselection) — standard (SS) then fullPath — plus comparisons to 2023 and to the previously-derived 2022 efficiencies.

Production workflow

1. **Data → TnP trees.** CMSSW analyzer over 2024 data via CRAB ⇒ per-event tag-and-probe ntuples (`tnpTree_2024.root`).

2. **hadd + η–R_9 reweighting.** Merge and attach the per-event η–R_9 weight ⇒ fitter-input trees `fitter_input_2024_{Seed,Unseed}_SS.root`.

3. **Tag-and-probe fits.** `egm_tnp_analysis` (CMSSW_10_6_8), driven by `run_HLT_2024_all.sh`: fit the m_{ee} spectrum, pass and fail, in each (η, R_9, E_T) bin ⇒ `egammaEffi.txt`.

4. **Consolidation.** Parse all `egammaEffi.txt` ⇒ `efficiencies_from_txt.csv`; build efficiency-vs-E_T plots, copy per-bin mass fits.

5. **Publish.** Push CSV, plots, fits to the GitHub repo; host the interactive explorer on GitHub Pages.

Dataset — 2024

Data = TnP probe photons; **MC** = gluon-fusion $H \to \gamma\gamma$ ($m_H = 90$) signal, used only to build the η–R_9 weight map.

	Data	**MC (for η–R_9 weights)**
sample	`tnpTree_2024.root`	`GluGluHto2G_Par-M-90` (RunIII2024Summer24)
tree	`diphotonTnP/tnpPhoHLT`	`photonPairTree/photonTree`
branches	`probe_full5x5_r9, probe_eta`	`full5x5_r9, eta`
events	159,861,517	2,413,586

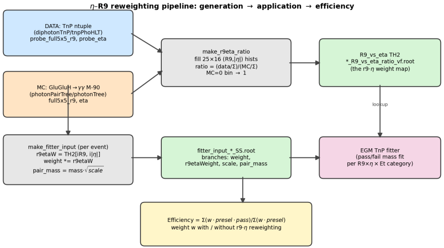
- MC dataset: `GluGluHto2G_Par-M-90_TuneCP5_13p6TeV_amcatnloFXFX-pythia8` (`RunIII2024Summer24MiniAODv6-150X_mcRun3_2024_realistic_v2-v2`).
- Weight map binning: 25 $R_9 \times 16$ $|\eta|$ bins; mean data-event weight ≈ 1.69.
- DY$\to$ *ee* MC was also produced but is *not* used — the weights come from the $H \to \gamma\gamma$ M-90 sample.
- Settings carry a placeholder $\mathcal{L} = 109.08\,\mathrm{fb}^{-1}$ (not shown as a confirmed luminosity on plots).

η–R_9 reweighting — what and why

- Probe photons from $Z \to ee$ do not have the same $(R_9, |\eta|)$ shape as $H \to \gamma\gamma$ signal photons.
- Build a data/MC ratio map in $(R_9, |\eta|)$ (density-normalised) and multiply each probe by its cell value:

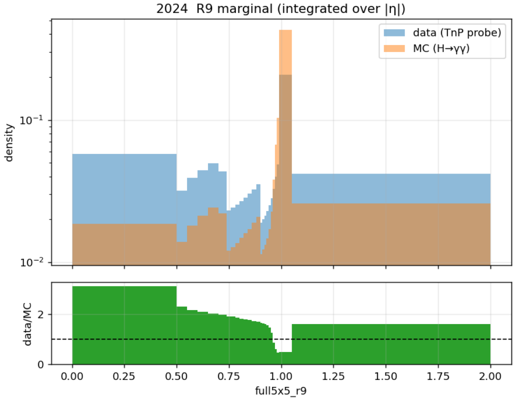
$$w_{ab} = \frac{N^D_{ab}/\Sigma N^D}{N^M_{ab}/\Sigma N^M}, \quad w_{ab} = 1 \text{ if } N^M_{ab} = 0.$$

- It is a per-event **multiplicative weight**, *not* a cut — it reweights the probe population, applied to both pass and fail.
- Two variants: `noPresel` (map from all probes) vs `withPresel` (map after the offline preselection).

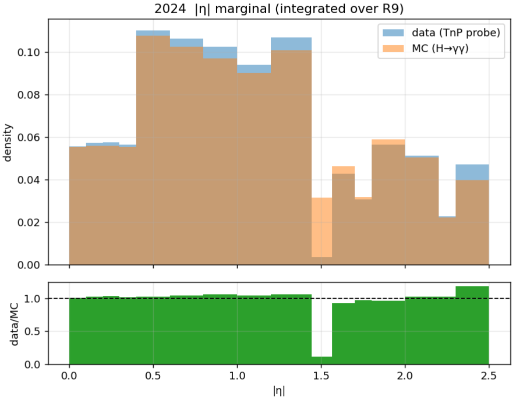


Generate map → apply per event → enters the TnP fit as a weight.

η–R_9 distributions that drive the weights (2024)



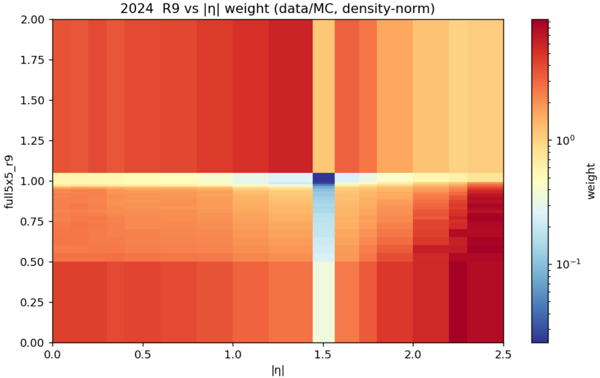
R_9 marginal (integrated over $|\eta|$), data vs MC + ratio.



$|\eta|$ marginal (integrated over R_9), data vs MC + ratio.

The reweighting acts most strongly at **low** R_9 and in the **endcaps**, where TnP data and $H \to \gamma\gamma$ MC differ most; the $|\eta|$ shapes agree well after density-normalisation (ratio ≈ 1).

η–R_9 2-D weight map (2024)



- Stored as a `TH2` (`R9_vs_eta`) in `2024_R9_vs_eta_ratio_vf.root`.
- Log colour scale, pivot at $w = 1$ (white).
- Red ($w \gg 1$): low-R_9 / high-$|\eta|$ probes are up-weighted.
- Blue band at $R_9 \approx 0.95$–1.0 and the grey EB–EE gap column (1.44–1.57) are common to all years.
- Same 25×16 binning as the marginals.

Preselection — exact base cuts

Every fit bin applies the same `cutBase`, plus a per-E_T-bin R_9 category cut and the leg "pass" flag. `noPresel` and `withPresel` use the **identical** cutBase — they differ only in *which* η–R_9 weight branch is applied.

```
   tag_Ele_pt > 35  &&  abs(tag_sc_eta) < 2.5  &&  tag_mva > -0.6
&& probe_pt > 30                 # seeded leg      (unseeded: probe_pt > 18)
&& ph_sc_abseta < 1.4442         # EB region       (EE: 1.566 < ph_sc_abseta < 2.5)
&& probe_mva > -0.9  &&  probe_hoe < 0.08
&& ph_full5x5x_r9 > 0.5          # region R9 floor
&& probe_sieie < 0.011           # EB              (EE: probe_sieie < 0.032)
&& probe_phoIso_rhoCorr < 3.0
&& probe_trkSumPtHollowConeDR03 < 6.0
&& ( ph_full5x5x_r9 > 0.8 || probe_pfRelIso03_chg_quad*probe_pt < 20
                          || probe_pfRelIso03_chg_quad < 0.3 )   # H->gg isolation OR-clause

# unseeded leg additionally:  && tag_seededPass == 1
# fullPath variant adds:      && probe_passFullPath == 1
# per-pt-bin R9 category cut (additionalCuts), e.g. EB 0.50-0.55:
#    ph_full5x5x_r9 > 0.50 && ph_full5x5x_r9 < 0.55
# pass flag (numerator):  probe_passSeed == 1   (unseeded: probe_passUnseed == 1)
```

TnP fitter — how the fits are run

CMS EGamma POG egm_tnp_analysis (CMSSW_10_6_8). For each category × leg the driver calls tnpEGM_fitter.py through a fixed sequence:

```
# run_HLT_2024_all.sh  <WEIGHT> <SET>   WEIGHT=noPresel|withPresel  SET=normal|fullPath
for CAT in EB_R9_0p50to0p55 EB_R9_0p55to0p85 EB_R9gt0p85 EE_R9_0p90to0p93 EE_R9gt0p93; do
 for LEG in seed unseed; do
   FITTER $CFG --flag $FLAG --checkBins     # sanity-check bin definitions
   FITTER $CFG --flag $FLAG --createBins    # build (ph_sc_abseta, ph_sc_et) bins
   FITTER $CFG --flag $FLAG --createHists    # fill pass/fail pair_mass hists (WEIGHTED)
   FITTER $CFG --flag $FLAG --doFit --bwDscb     # BWxDSCB signal + RooCMSShape bkg (nominal)
   FITTER $CFG --flag $FLAG --doFit --bwDscbExp   # BWxDSCB signal + Exponential bkg (
     alternate)
   FITTER $CFG --flag $FLAG --sumUp          # write egammaEffi.txt (eff, +err, -err per bin)
 done
done
```

--createHists fills, per (η, E_T) bin, a *passing* and a *failing* m_{ee} histogram (pair_mass, 80 bins, $[50, 130]$ GeV) with the per-event weight; "passing" $=$ probe_passSeed==1 (or probe_passUnseed==1).

Reweighting + preselection inside the fitter

The η–R_9 weight is looked up per event, written as a branch in the input tree, selected by the settings file, and applied by the fitter when it fills the mass histograms:

```
# (1) per-event lookup (make_fitter_input_2024): each probe -> one weight
ix = clip(digitize(r9,       R9_EDGES ) - 1, 0, nR9 -1)   # 25 R9 bins
iy = clip(digitize(abs(eta), ETA_EDGES) - 1, 0, nEta-1)   # 16 |eta| bins
r9etaWeight = weights_2d[ix, iy]
weight_{no,with}_presel = weight0 * r9etaWeight           # branch in fitter_input_2024_*.
     root

# (2) the settings file SELECTS which variant to apply:
tnpSample('data_2024_noPresel_...', _input, lumi=109.08, weight='weight_no_presel')
#   withPresel -> weight='weight_with_presel'

# (3) the fitter APPLIES it filling the pass/fail mass histograms (rootUtils.py):
cutPass = '( %s && %s ) * %s' % (cuts,    flag, sample.weight)   # weighted PASS
cutFail = '( %s && %s ) * %s' % (cuts, notflag, sample.weight)   # weighted FAIL
```

The *same* weight enters both pass and fail templates, so ε is reweighted consistently.

`noPresel`/`withPresel` differ only in the map the weight came from; the preselection is the `cutBase` applied identically in both.

Fit model — functional forms

For each (η, R_9, E_T) bin the m_{ee} spectrum of tag–probe pairs is fit in $[50, 130]$ GeV, separately for probes that *pass* and *fail* the leg requirement:

$$\varepsilon = \frac{N^{\text{sig}}_{\text{pass}}}{N^{\text{sig}}_{\text{pass}} + N^{\text{sig}}_{\text{fail}}}\,.$$

- **Signal:** Breit–Wigner $\otimes$ double-sided Crystal Ball (**bwDSCB**) — a relativistic Z Breit–Wigner convolved with a two-sided Crystal-Ball resolution function.
- **Background (nominal, `bwDscbCms`): RooCMSShape** $=$ error-function $\times$ exponential.
- **Background (alternate, `bwDscbExp`): Exponential** — drives the background systematic (present in every CSV and in the explorer).

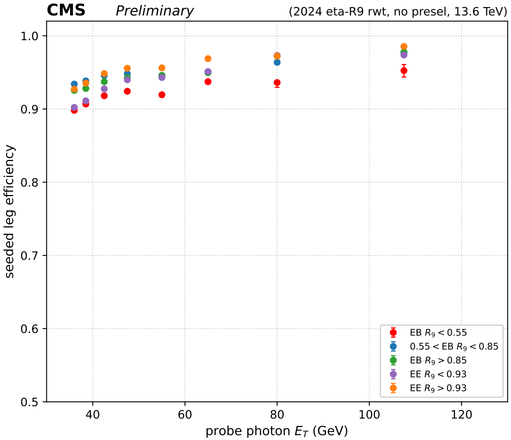
Quoted uncertainties σ_+/σ_- are the asymmetric fit (statistical) errors on ε; the two background models give the nominal value and the background systematic.

Fit parameters, ranges and bounds

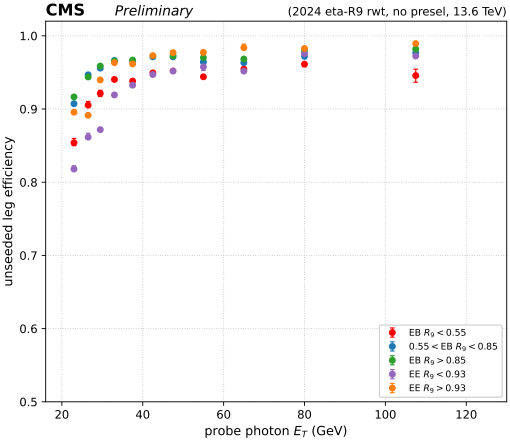
Signal-model parameter initialisation `[start, low, high]` (pass P, fail F):

```
# tnpParBWDSCBFit  (signal = Breit-Wigner (x) double-sided Crystal Ball)
meanP[0.0,-3.0,3.0]   sigmaP[1.5,0.3,5.0]    meanF[0.0,-3.0,3.0]   sigmaF[1.5,0.3,5.0]
alpha1P[1.5,0.5,5.0]  n1P[2.0,1.0,10.0]      alpha2P[1.5,0.5,5.0]  n2P[2.0,1.0,10.0]
alpha1F[1.5,0.5,5.0]  n1F[2.0,1.0,10.0]      alpha2F[1.5,0.5,5.0]  n2F[2.0,1.0,10.0]

# nominal background = RooCMSShape (erf x exp):
acmsP[60.,50.,100.]  betaP[0.05,0.01,0.08]  gammaP[0.1,0,1]  peakP[90.0]   # + acmsF/betaF/
     gammaF/peakF

# alternate background (tnpParBWDSCBExpFit) replaces CMSShape with an exponential:
alphaP[0.,-2.,1.]    alphaF[0.,-2.,1.]

# fit variable: pair_mass, 80 bins in [50,130] GeV
# efficiency = N_sig_pass / (N_sig_pass + N_sig_fail)
```

Results — efficiency vs E_T (standard SS)



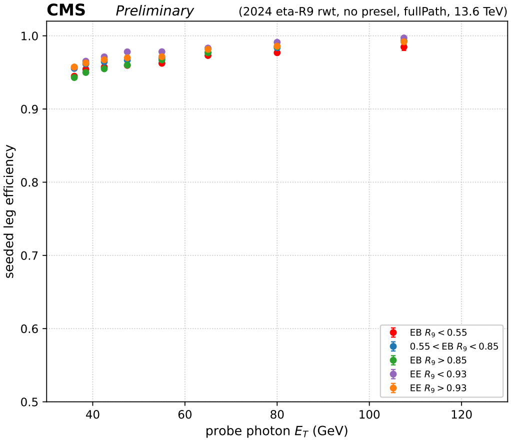
Seeded (lead) leg.

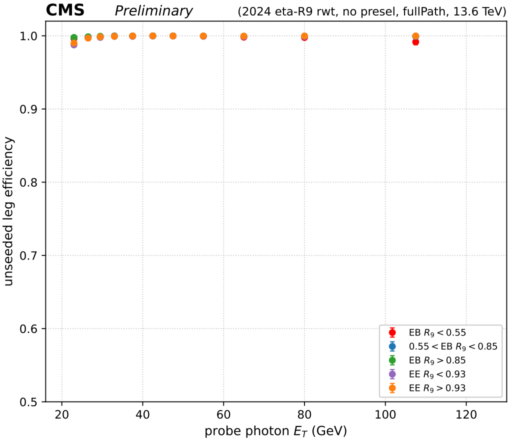


Unseeded (sublead) leg.

η–R_9-reweighted, standard track-isolation selection; nominal bwDSCB $\otimes$ RooCMSShape fit; five η/R_9 categories overlaid.

Results — efficiency vs E_T (fullPath)

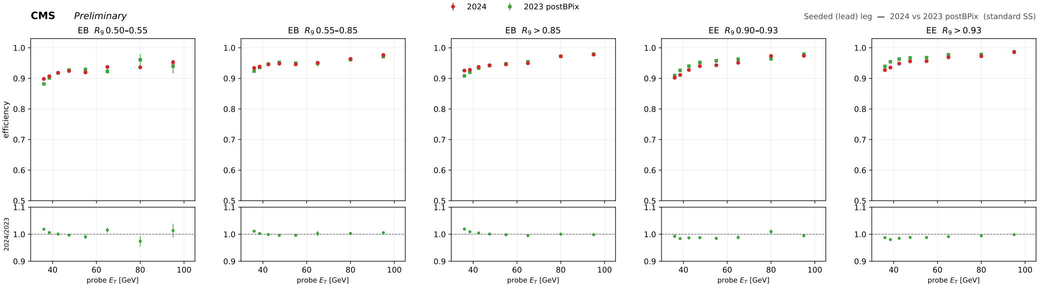




Seeded (lead) leg, fullPath. Unseeded (sublead) leg, fullPath.

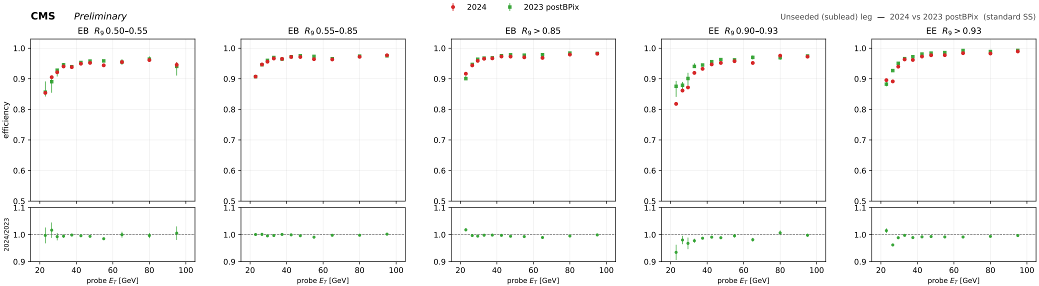
Same variant with the full HLT-path requirement (`probe_passFullPath`$= 1$) added on top of the standard selection.

2024 vs 2023 — seeded leg (standard SS)



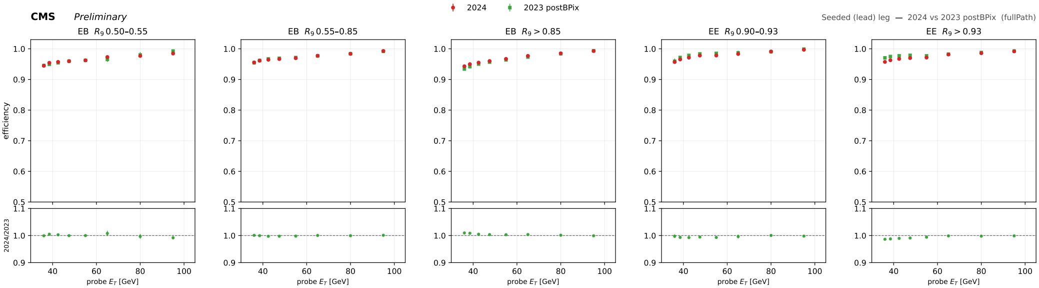
Per-category efficiency vs E_T: **2024** overlaid with **2023 postBPix**; bottom row is the ratio 2024/2023. Built from the CSV values that back the result repos. Agreement within a few percent across all five η/R_9 categories.

2024 vs 2023 — unseeded leg (standard SS)



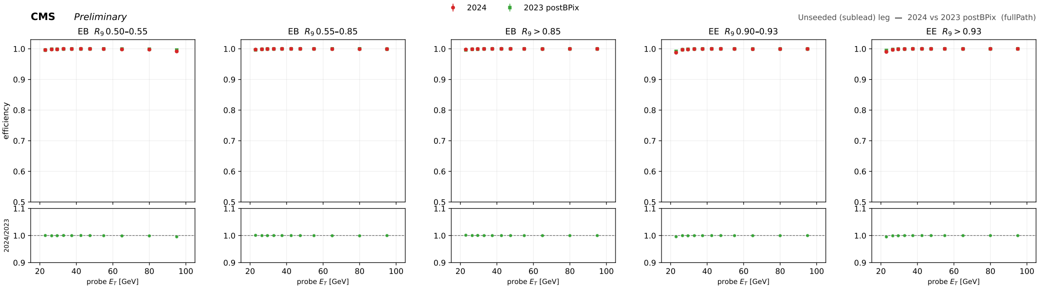
Unseeded (sublead) leg; same overlay + ratio to 2023 postBPix.

2024 vs 2023 — seeded leg (fullPath)



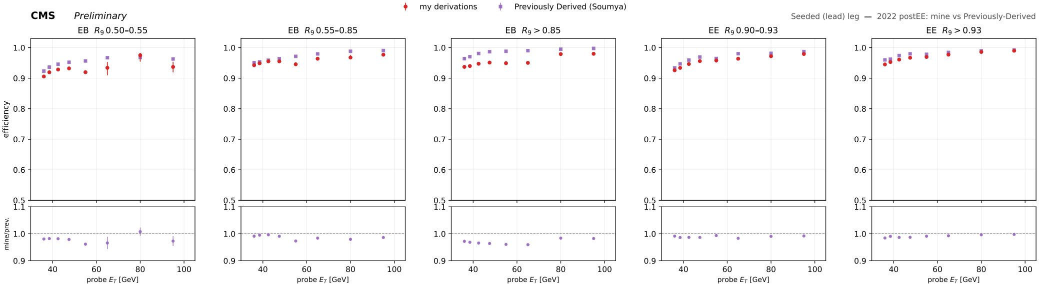
fullPath selection: **2024** vs **2023 postBPix**, seeded leg.

2024 vs 2023 — unseeded leg (fullPath)



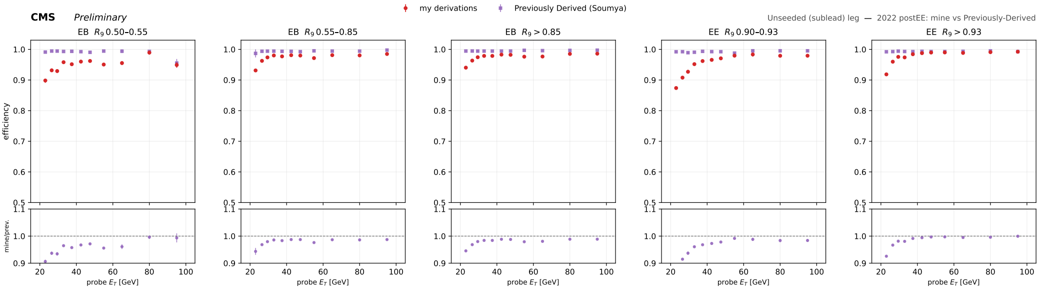
fullPath selection: **2024** vs **2023 postBPix**, unseeded leg.

2022 — mine vs previously derived (seeded)



My derivations (bwDSCB fitter) vs the **previously-derived** 2022 postEE efficiencies (Soumya, correctionlib); ratio mine/previous below. Seeded leg.

2022 — mine vs previously derived (unseeded)



Unseeded leg. Full interactive cross-year overlay — drag/click any curve, hover a point for its mass fit — at the **GitHub Pages explorer**.

Summary

- First 2024 low-mass diphoton HLT leg efficiencies: tag-and-probe, η–R_9 reweighted, both legs, five η/R_9 categories, four variants (`noPresel`/`withPresel` $\times$ standard/fullPath).
- η–R_9 reweighting built from a 25×16 data/MC map ($H \rightarrow \gamma\gamma$ M-90 MC), applied as a per-event weight to the mass templates — not a cut.
- Signal bwDSCB $\otimes$ {RooCMSShape (nominal), Exponential (alt)}; efficiency from fitted pass/fail signal yields; asymmetric statistical errors.
- Results published; cross-year overlay in the interactive explorer.

Repo: HLT-diphoton-efficiency-2024-results · Explorer: santanumahata.github.io/HLT-diphoton-efficiency-2024-results

Full per-bin values + method: `HLT_efficiency_production_2023_2024.pdf`

Backup

Cut-and-count cross-check: weighted pass-fraction (no background subtraction, no fit), overlaid on the nominal fit — same categories, same E_T bins.

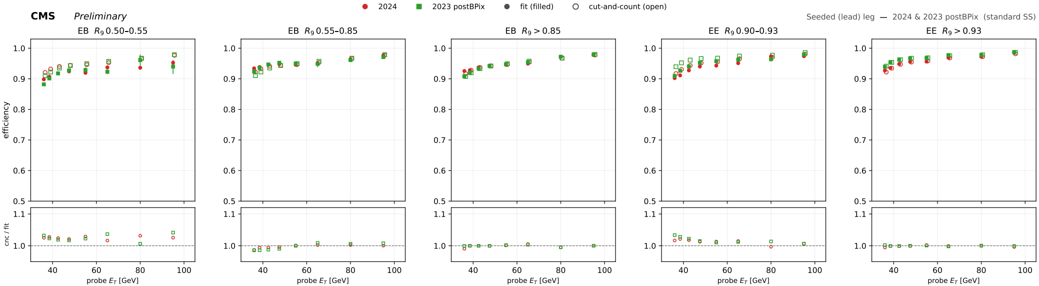
Backup — cut-and-count definition

Weighted pass-fraction on the same `cutBase` used by the fit, no signal/background model:

```
# per (leg, category, ET bin): same cutBase (+ passFullPath for the fullPath variant),
# same eta-R9 weight branch as the fit, pair_mass in [50,130] GeV (the fit's own mass window)
sel = eval(cutBase) & (50 < pair_mass < 130)
eff_cnc = Sum( weight[sel & pass] ) / Sum( weight[sel] )     # pass = probe_passSeed/Unseed
     == 1
```

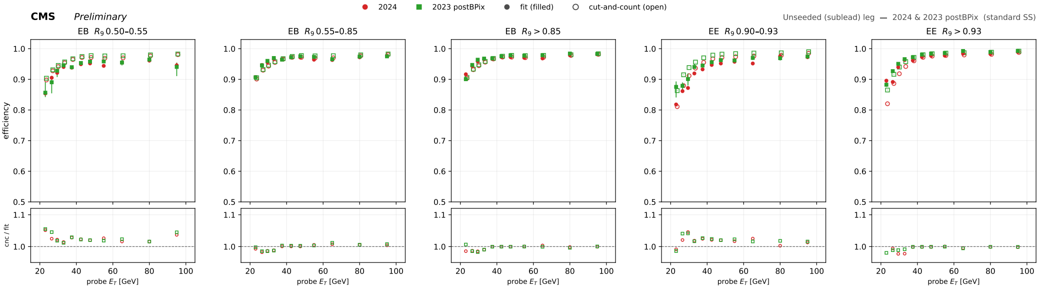
No signal/background separation — so cnc sits slightly *above* the fit (uncut background under the Z peak is not subtracted). Used only as an independent, model-free sanity check on the fitted efficiencies; **not** a competing measurement.

Backup — 2024 vs 2023 postBPix, standard SS (seeded)

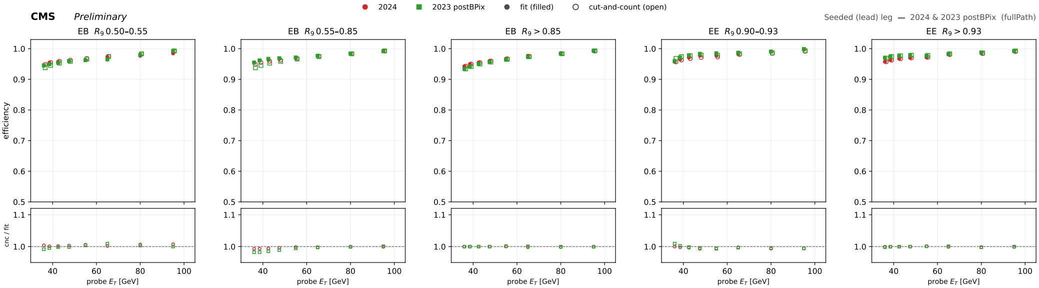


Filled = fit, open = cut-and-count; bottom row is the ratio cnc/fit.

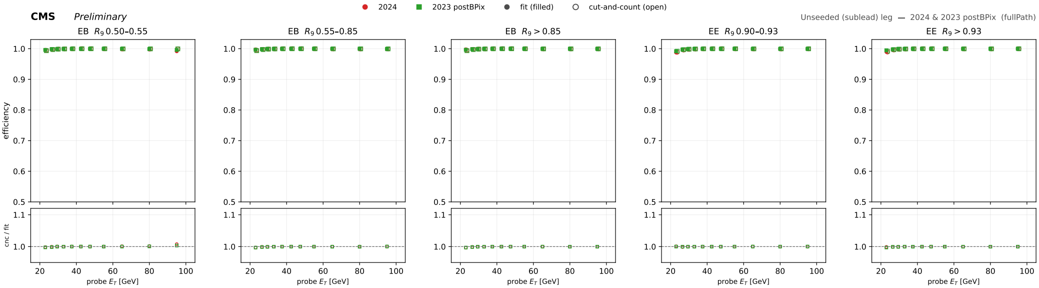
Backup — 2024 vs 2023 postBPix, standard SS (unseeded)



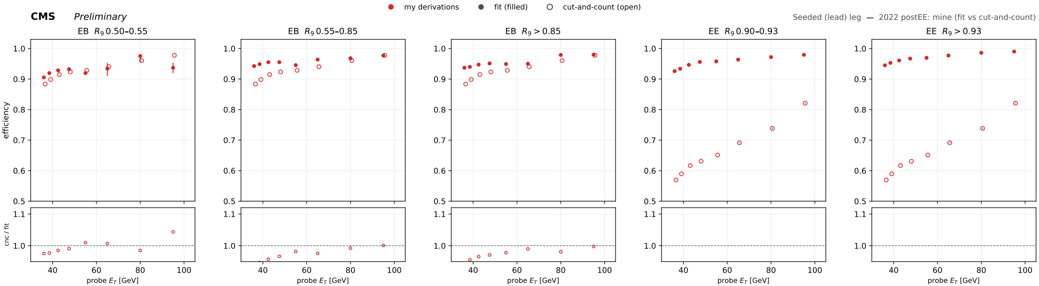
Backup — 2024 vs 2023 postBPix, fullPath (seeded)



Backup — 2024 vs 2023 postBPix, fullPath (unseeded)



Backup — 2022 postEE mine, fit vs cut-and-count (seeded)



Backup — 2022 postEE mine, fit vs cut-and-count (unseeded)